

Destination-Based EV Charging Infrastructure in India

A Business Model and Feasibility Study for Converting Existing Destination Parking into Distributed EV Charging Infrastructure



Cover: electric vehicles charging in destination parking during a routine visit. Source: commissioned illustration for this report.

ITEM	DETAIL
RESEARCH QUESTION	Under what conditions can existing destination parking be converted into economically viable EV charging infrastructure in India?
METHOD	Secondary market research · Unit-economics modelling · Scenario & sensitivity analysis (no primary data collected)
DATE	September 2026
VERSION	1.0

This document is an academic business research report — a feasibility investigation, not a validated business plan. It does not constitute financial advice or an investment recommendation. All economic models are directional and contain clearly labelled assumptions requiring empirical validation.

ABSTRACT

Abstract — Research Summary

Background. India's electric vehicle market set new records in 2025. Total EV retail sales reached 2.27 million units, up 16% year-on-year (Autocar Professional, 2026), and electric passenger vehicle sales grew 77% to 176,538 units (Autocar Professional, 2026). Public charging infrastructure expanded nearly sixfold in three years to 29,277 stations by August 2025 (Ministry of Power, 2025). Yet the charger-to-EV ratio stands at roughly 1:235 against a global benchmark of 1:6 to 1:20 (The Economic Times, 2025), and only 22,753 of 27,737 installed stations were operational as of March 2026 — an 18% non-functional rate (MMCM, 2026).

Problem. Dedicated charging sites face land costs, capital intensity and grid-connection constraints; premium urban leases can absorb up to 40% of project budgets (Mordor Intelligence, 2025). Meanwhile, thousands of commercial destinations — malls, hotels, hospitals, offices, universities — already own parking, customer dwell time, footfall and electricity connections. Whether these existing assets can be integrated into a viable destination-charging model is the central problem this study investigates.

Research objective. To determine the conditions under which existing destination parking can be converted into economically viable EV charging infrastructure in India — evaluating the model critically, separating what is supported by evidence from what remains assumption or hypothesis.

Methodology. The study combines secondary market research (regulatory data, Vahan registrations, industry reports) with original unit-economics modelling, scenario analysis and sensitivity testing. No customer interviews, host negotiations or pilot data have been collected; this absence is disclosed throughout and every major claim is classified as FACT, ESTIMATE, ASSUMPTION or HYPOTHESIS.

Major findings. The market opportunity is real but narrow: the addressable pool is concentrated among urban EV owners without reliable home charging, directionally estimated at 100,000–150,000 passenger EVs. At reference-case inputs (Rs.12/kWh price, Rs.9/kWh tariff, 20% host share), a charging bay cannot break even at any plausible utilisation — the operator retains only Rs.0.60/kWh, implying a breakeven of ~30 sessions/day. Viability emerges only under stricter parameters (Rs.15 price, Rs.8 tariff, 15% share), where breakeven falls to ~3.4 sessions/bay/day with ~5.5-year payback. Site utilisation as low as 5% has been observed at deployed public stations (MMCM, 2026), and large operators — Tata Power, Jio-bp, Statiq, Bolt.Earth — are expanding aggressively.

Business implications. The destination model structurally avoids land-acquisition cost and is aligned with Ministry of Power hosting guidelines. It is conditionally viable: site selection, tariff discipline and utilisation above ~3.5 sessions/bay/day are non-negotiable. A structured 3–5 site pilot is the recommended next step before any scaling commitment.

Key limitations. Driver willingness to use destination charging, host willingness to allocate parking, and any incremental host spending uplift are all unvalidated hypotheses. All financial figures are directional modelling, not forecasts.

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Evidence classification. Throughout this report all major claims are classified as: **[FACT]** — sourced from verifiable data; **[ESTIMATE]** — derived from available evidence; **[ASSUMPTION]** — modelling input requiring validation; **[HYPOTHESIS]** — proposition requiring empirical testing.

Reading guide. Chapters 1-2 establish context and the research gap. Chapters 3-4 define the business model and the customer. Chapter 5 contains the core economic analysis, including a corrected reference case and a viability frontier. Chapters 6-7 cover strategy, validation and risk. Chapter 8 states what is proven, what is not, and what must be tested next.

CHAPTER 01

01 — Introduction

1.1 Background

[FACT] India's EV industry crossed the 2-million annual sales milestone for the first time in November 2025 and closed CY2025 at 2.27 million units, up 16% on CY2024's 1.95 million (Autocar Professional, 2026). Electric passenger vehicles posted their strongest year on record: 176,538 units, up 77% year-on-year, with 11 months above 11,000 units and a festive peak of 19,161 units in October (Autocar Professional, 2026). The last three calendar years account for the large majority of all EVs ever sold in India, indicating accelerating — not linear — adoption momentum.

[ESTIMATE]

[FACT] Public charging infrastructure has grown nearly sixfold in three years: from 5,151 stations in December 2022 to 25,202 in December 2024 and 29,277 by August 2025, operated by 83 charge point operators (Ministry of Power, 2025, via ORF, 2025). Yet scale remains the binding constraint. India has roughly one public station per 235 EVs versus a global benchmark of one per 6-20 (The Economic Times, 2025), and reaching the government's 30% EV penetration ambition for 2030 would require an estimated 1.32 million public stations — more than 40 times the current base (ORF, 2025).

[FACT] Policy direction is explicitly supportive. Setting up charging stations is a de-licensed activity under Ministry of Power guidelines; the PM E-DRIVE scheme allocates Rs.2,000 crore toward 72,300 public chargers; and official guidance identifies malls, workplaces, fuel stations and residential complexes as preferred hosting sites (Avaada, 2026; Bolt.Earth, 2025). The destination-charging model evaluated here sits directly inside this policy corridor.

1.2 Problem Statement

Vehicle growth and usable charging supply are diverging. Dedicated charging infrastructure must be located where charging is useful, but commercially attractive urban land is expensive — business-district leases in Mumbai consume up to 40% of project budgets and extend payback periods (Mordor Intelligence, 2025) — while grid-connection upgrades add further capital and delay. Operational quality compounds the problem: an 18% national non-functional rate, up to 60% on the worst networks, and site utilisation as low as 5% at many deployed stations (MMCM, 2026).

Simultaneously, thousands of destination businesses already possess the three assets a charging operator must otherwise buy or build: parking inventory, customer dwell time, and electricity connections — plus existing footfall. The underlying problem is therefore allocative: whether these dormant assets can be integrated into a destination-charging network that is economically sustainable at the level of the individual site, not merely attractive in aggregate market statistics.

1.3 Research Objectives

- **01.** To assess the scale and composition of India's EV market relevant to destination charging.
- **02.** To define the criteria that make a destination site suitable for EV charging conversion.
- **03.** To model the unit economics of a single destination-charging bay under multiple utilisation scenarios.
- **04.** To evaluate whether host businesses can receive sufficient economic incentive to allocate parking.
- **05.** To identify the assumptions and hypotheses requiring empirical validation before any scaling commitment.

1.4 Research Questions

NO.	RESEARCH QUESTION
RQ1	What portion of India's EV market is behaviourally and technically relevant to destination charging?
RQ2	What destination characteristics — dwell time, parking, electricity capacity, EV density — determine site suitability?
RQ3	At what utilisation rate does a destination-charging bay break even, and how sensitive is this to price, tariff and revenue share?
RQ4	Can host businesses receive sufficient economic value to justify allocating parking bays to EV charging?
RQ5	What hypotheses must be validated through primary research and a pilot before the model can be responsibly scaled?

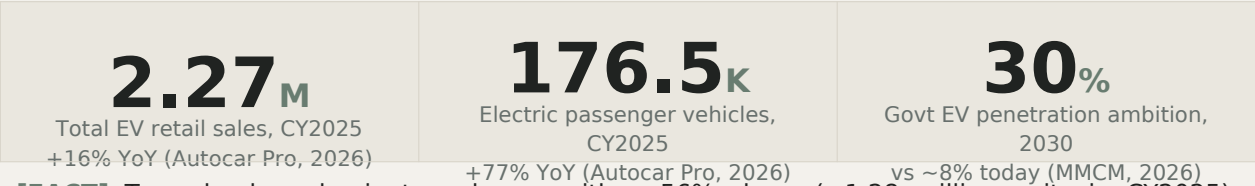
1.5 Scope

WITHIN SCOPE	OUTSIDE SCOPE
Passenger EV segment (four-wheelers) Urban and Tier-1 Indian markets AC destination chargers (7–22 kW) Commercial destination categories Operator-host partnership model Unit economics and scenario modelling	Highway DC fast charging Two- and three-wheeler charging Rural / Tier-3 markets Battery swapping infrastructure Fleet depot charging Primary interviews (proposed, not conducted)

CHAPTER 02

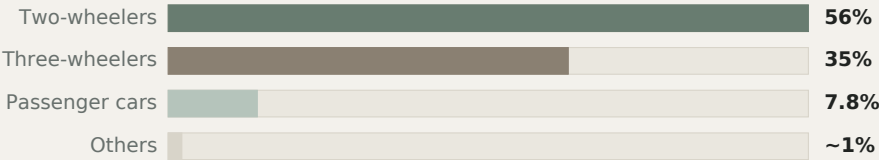
02 — Market & Literature Review

2.1 Indian EV Market



[FACT] Two-wheelers dominate volumes with a 56% share (~1.28 million units in CY2025), followed by three-wheelers at ~35%; together they account for 91% of EV sales (Autocar Professional, 2026). Passenger four-wheelers are the smallest segment by units but among the fastest-growing in percentage terms, and — critically for this study — the segment whose range, charge duration and destination dwell patterns align best with AC destination charging.

Figure 2.1 — India EV Sales Composition by Vehicle Type, CY2025



Source: Autocar Professional analysis of Vahan data, January 2026. Note: CY2025 figures; e-2W 56%, e-3W ~35%, e-PV ~7.8%.

[FACT] Tata Motors remains the largest player in the electric passenger vehicle segment (Cornell University, 2025). The segment’s trajectory — 99,693 units in CY2024 to 176,538 in CY2025 — implies that a destination-charging operator entering today builds ahead of mass adoption: an early-mover timing opportunity paired with early-period utilisation risk.

2.2 EV Charging Behaviour

Charging behaviour decomposes into five location types with distinct economics:

LOCATION	USER TYPE	ECONOMICS	RELEVANCE TO THIS MODEL
Home	Owners with parking	Cheapest per kWh; slow	Low — self-sufficient
Workplace	Employees	Often subsidised; long dwell	Medium — offices are viable hosts
Fleet / depot	Fleets, ride-hailing	High volume, managed	Medium — utilisation anchor
Public fast (DC)	En-route travellers	Premium price, fast turn	Low — different model
Destination (AC)	Shoppers, diners, guests	Value aligned with dwell	High — this study’s subject

[FACT] Home charging dominates energy delivery wherever it is available, at Rs.5-8/kWh versus Rs.15-30/kWh on public networks (EVblogs, 2025). The addressable pool for destination charging is therefore concentrated among owners who cannot reliably charge at home — apartment dwellers, households without dedicated parking, ride-hailing drivers and long-trip urban drivers. Fleet size must never be conflated with public-charging demand.

2.3 Destination Charging — Definition and Rationale

Destination charging is infrastructure installed where users arrive for a primary purpose other than charging — shopping, dining, medical care, work, overnight stay. The session is incidental to the visit; the charger monetises time the user would have spent at the location regardless. Its economics rest on dwell-time alignment: a 7-22 kW AC charger can add 30-50 km of range over a 2-3 hour visit, sufficient for most urban use cases.

The model is structurally distinct from both home charging (cheapest, private) and highway fast charging (premium, throughput-driven). It competes on convenience and routine integration — charging that requires no detour, no wait and no dedicated trip. That convenience must be worth the premium over home tariffs; whether drivers agree is Hypothesis H1 (Chapter 7).

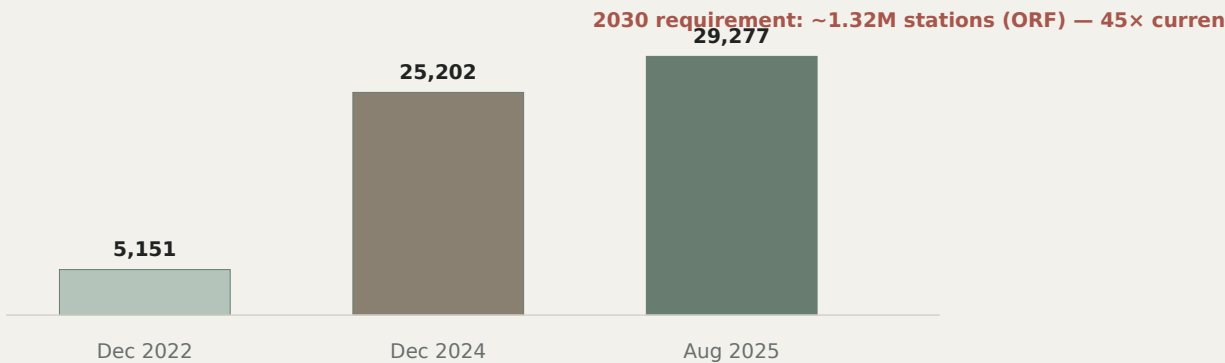


Figure 2.4 — Destination parking as convertible infrastructure: marked bays, existing electrical access and captive dwell time. Source: commissioned illustration for this report.

2.4 Existing Research / Industry Evidence

INFRASTRUCTURE EXPANSION	OPERATIONAL PERFORMANCE
[FACT] Stations grew 5,151 (Dec 2022) → 25,202 (Dec 2024) → 29,277 (Aug 2025) across 83 CPOs (Ministry of Power, 2025). [FACT] Concentration is severe: five states hold ~60% of stations; Bengaluru alone reports ~5,880 (ORF, 2025; Avaada, 2026). [FACT] Including semi-public points, the count crosses ~39,500 chargers by late 2025 (Bolt.Earth, 2026).	[FACT] Of 27,737 stations reported to Parliament, only 22,753 were operational — 18% non-functional; BPCL at 60%; utilisation as low as 5% at many sites (MMCM, 2026). [FACT] Premium urban leases consume up to 40% of project budgets (Mordor Intelligence, 2025). [FACT] Policy favours land-sharing: malls, workplaces and fuel stations are designated preferred hosts (Bolt.Earth, 2025).

Figure 2.2 — Public Charging Stations in India, Dec 2022 - Aug 2025



Source: Ministry of Power (2025) via ORF (2025); 2030 requirement per ORF. The 2030 bar is 45× the current base and shown as annotation, not to scale.

2.5 Research Gap

Market data establishes EV and infrastructure growth. It does not establish whether individual destination sites can achieve economically sustainable utilisation.

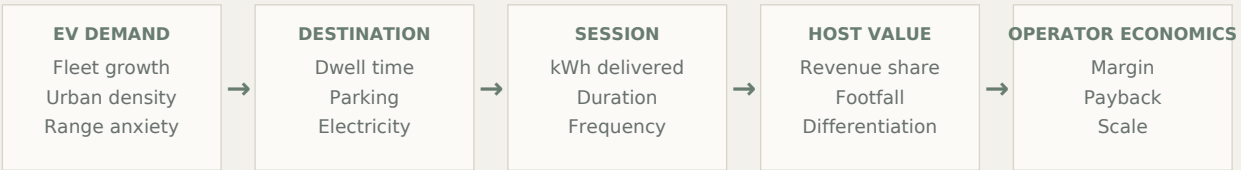
Published utilisation spans from below 5% at weak sites to far higher at mature ones; the distribution is unknown. Site-level session volume requires primary data absent from existing literature.

This gap defines the study’s contribution: it translates macro demand signals into site-level economics, identifies the exact thresholds that must hold, and specifies how each threshold will be tested — rather than asserting viability from aggregate growth rates.

2.6 Conceptual Framework

The analytical framework connects macro EV demand through the site-selection filter to session-level economics:

Figure 2.3 — Destination Charging Analytical Framework



Source: author’s conceptual framework. Each arrow is a hypothesis: demand must convert to arrivals, arrivals to sessions, sessions to host value, and host value to sustainable operator margin.

CHAPTER 03

03 — Proposed Business Model

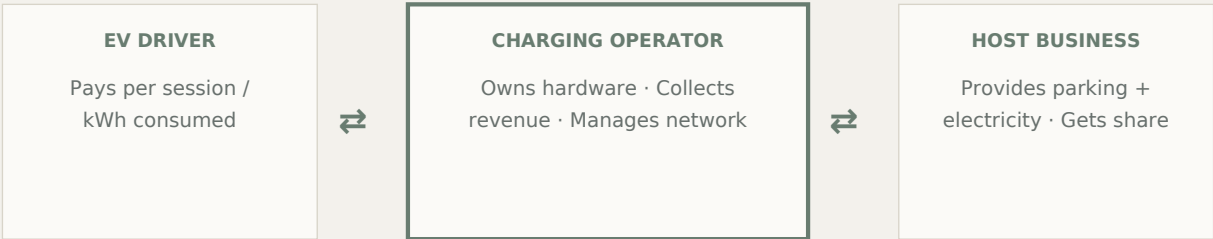
3.1 Concept

An independent charging operator partners with commercial destination businesses. The operator supplies, installs, operates and maintains AC charging equipment in selected bays of the host’s existing parking; the host provides land, electricity access and footfall. Charging revenue is shared under a pre-agreed commercial structure. The core insight is asset synthesis: the operator gains location access without land acquisition; the host monetises underutilised parking without operational complexity. Neither party captures the full value alone.

The model targets AC charging (7-22 kW) because destination dwell times of 45 minutes to several hours match AC charge curves, equipment and grid-connection costs are an order of magnitude below DC fast charging, and most urban top-up needs are satisfied within a single visit.

3.2 Stakeholders

Figure 3.1 — Three-Party Commercial Structure



Source: author’s framework. The operator is the commercial hub: it contracts with drivers for energy and with hosts for site access.

Three contracts define the system: driver-operator (price per kWh, payment, reliability expectation), operator-host (revenue share or access fee, bay availability, contract term, branding), and the implicit driver-host relationship (the visit itself). Failure in any one contract — distrusted chargers, defecting hosts, absent drivers — collapses site economics.

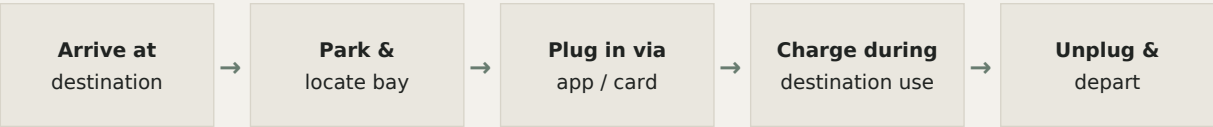
3.3 Three-Sided Value Model

EV DRIVER	HOST BUSINESS	OPERATOR
Charges during existing visit — no detour Predictable location in daily routine Reduced urban range anxiety AC speed matched to dwell [FACT] Range anxiety is a documented adoption barrier	Revenue share per session No capital or operational burden Better use of selected bays EV-friendly differentiation [ASSUMPTION] Incremental customer spend requires validation	Session revenue Location access without land cost Network scale via partnerships Software and data value [HYPOTHESIS] Scale creates a defensible position

Each column’s evidence status differs deliberately. Driver convenience is supported by charging-behaviour literature; host upside beyond the revenue share is assumed; operator defensibility is hypothesised and directly challenged by incumbent scale (Chapter 6).

3.4 Customer Journey

Figure 3.2 — Destination Charging Customer Journey



Source: author’s conceptual framework. Friction at any step — occupied bays, app failures, opaque pricing — converts directly into lost sessions.

Journey reliability is the product. With a national non-functional rate of 18% (MMCM, 2026), a new operator’s credible promise is not more chargers but dependable ones: live bay availability, single-tap payment, enforced EV-only bays and published uptime. Chapter 7 sets the uptime hypothesis (H6) at $\geq 90\%$.

3.5 Revenue Model

[FACT] The operator charges drivers per kWh. Observed public AC rates cluster around Rs.8-15/kWh (Pulse Energy, 2025), while full public-network tariffs span Rs.15-30/kWh (EVblogs, 2025). The operator pays electricity at the commercial tariff (Rs.6-15/kWh depending on state and load) plus the host’s share. The spread between consumer price and total delivered cost determines contribution per session — modelled rigorously in Chapter 5.

Three commercial structures are available; the choice trades host income certainty against operator risk:

STRUCTURE	MECHANICS	WHEN IT FITS
Revenue share	Host receives 15-25% of session revenue	Operator confident in utilisation; host accepts variability
Fixed access fee	Host receives fixed monthly rent per bay	Host demands certainty; operator carries utilisation risk
Hybrid	Base fee + reduced variable share	Early sites: de-risks host while preserving upside alignment

[ASSUMPTION] The hybrid structure is the recommended default for pilot sites: it secures host commitment during the unproven ramp-up period while keeping both parties exposed to session volume.

CHAPTER 04

04 — Market & Customer Analysis

4.1 Target Customer Segments

SEGMENT	CHARACTERISTICS	RELEVANCE	STATUS
Apartment-dwelling owners	No home charging; regular destination pattern	High — primary pool	[HYPOTHESIS]
Villa / bungalow owners	Reliable home charging	Low — supplementary only	[FACT]
Ride-hailing / fleet drivers	High daily km; no employer charging	Medium-High — frequent users	[HYPOTHESIS]
Long-trip urban drivers	Opportunity charging mid-journey	Medium	[HYPOTHESIS]
Corporate employees	Office-adjacent; possible subsidy	Medium — model-dependent	[ASSUMPTION]

Segmentation matters because willingness to pay a premium over home tariffs concentrates in the first and third rows. Marketing spend aimed at villa owners with home chargers would subsidise sessions that add no network value — a customer-acquisition discipline the pilot must enforce (Appendix D).

4.2 Destination Category Assessment

DESTINATION	TYPICAL DWELL	RELEVANCE	HOST CONSIDERATIONS	OPPORTUNITY
Mall	2–4 hrs	High	Large inventory; footfall data	High
Hotel (overnight)	8–14 hrs	Very high	Ideal overnight AC; premium fit	High
Office	7–9 hrs	High	Captive daily users; subsidy potential	High
University	4–8 hrs	High	Growing faculty EV use; policy support	High
Hospital	1–6 hrs	Medium-High	Variable visitor mix	Medium-High
Theatre	2–3 hrs	Medium-High	Peak windows; episodic footfall	Medium
Supermarket	30–90 min	Medium	Short dwell caps kWh; high turnover	Medium
Restaurant	45–90 min	Medium	Limited parking; high opportunity cost	Low-Medium

Table 4.1 — Destination Category Assessment. Source: author’s analysis. Dwell times are estimates absent primary research; opportunity ratings are qualitative judgements, not scored metrics.

The ranking logic is dwell × parking headroom × EV-owner affinity. Overnight and workday sites dominate because long dwell converts directly into delivered kWh per session — the revenue unit. Restaurants rank lowest despite high footfall because parking scarcity raises the host’s opportunity cost above any plausible revenue share.

4.3 Customer Pain Points

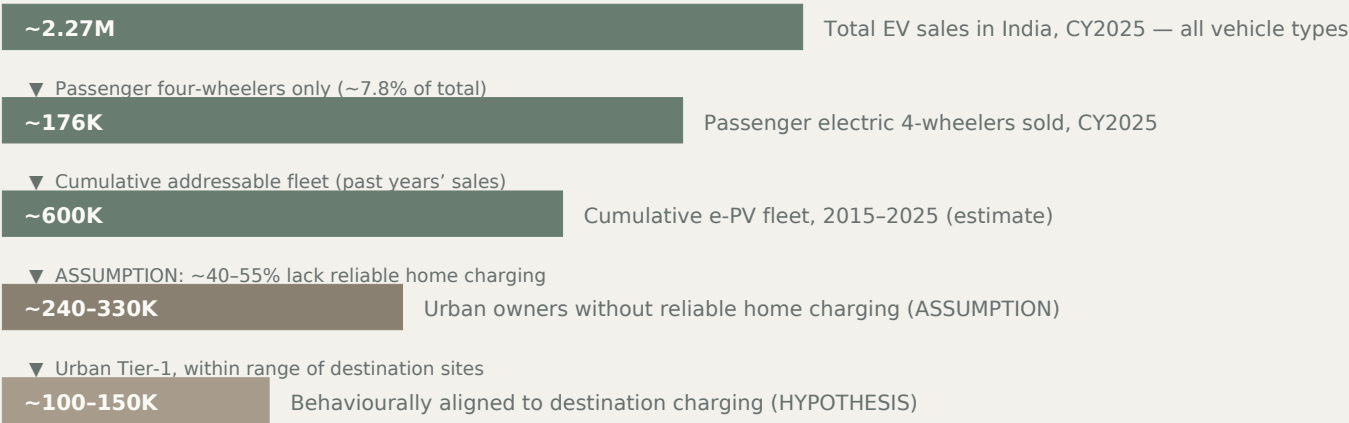
DISCOVERY & ACCESS	SESSION EXPERIENCE
— Range anxiety: depletion fear before reaching a charger — Availability: occupied or dead chargers on arrival — Detour time: diversion to dedicated stations — Queues: unpredictable waits at popular points	— Reliability: 18% national non-functional rate (MMCM, 2026) — Payment friction: fragmented apps and wallets — ICE-blocking: non-EVs occupying EV bays — Price opacity: tariffs vary by state, operator, hour

Every pain point is a product requirement: live availability, enforced bays, interoperable payment and published tariffs. Notably, reliability and ICE-blocking are operator-controllable — the two cheapest differentiators against incumbents.

4.4 Addressable Market — Filtering Framework

[ESTIMATE] The funnel below filters from total EV sales to the behaviourally relevant pool. All intermediate figures are directional estimates, not a market-sizing forecast.

Figure 4.1 — Addressable Market Funnel (directional, not a forecast)



Source: author’s analysis based on Autocar Professional (2026), IESA (2025) and Cornell University (2025). The 40–55% home-charging-gap assumption is the funnel’s pivotal input and is currently unvalidated.

The passenger-EV addressable pool is currently small but growing ~77% annually. At 176,538 new electric passenger vehicles in CY2025, the fleet compounds fast. An operator entering today builds ahead of mass adoption — timing opportunity paired with early-period utilisation risk.

[ASSUMPTION] A widespread unstated assumption in charging business cases is that EV owners will use public charging regularly. Home-charging preference challenges this directly: the segment that cannot charge reliably at home is the primary market, and its size in urban India cannot be determined from public data.

Evidence currently unavailable / requires primary validation: what share of India’s electric passenger-vehicle owners lack reliable home-charging access? This single input dominates the demand model and is the highest-priority question for Phase-1 interviews (Appendix A).

CHAPTER 05

05 — Economic Feasibility

Utilisation is the central economic variable — but price, tariff and revenue share decide whether any utilisation level suffices.

A bay with insufficient session volume fails regardless of EV growth. As shown below, under reference-case inputs the bay fails at every plausible utilisation level; viability requires a stricter parameter envelope defined in Section 5.3.

This chapter models a single AC bay (7-22 kW), corrects the reference case arithmetically, derives the breakeven frontier across parameter combinations, and defines the viable-parameter set used for the three-year scenario. All inputs are sourced from industry data where available and otherwise labelled.

5.1 Unit Economics — Single Charging Bay

CAPEX ITEM	RANGE (RS.)	STATUS
AC charger hardware (7-22 kW)	40,000 – 1,20,000	[FACT] Pulse Energy (2026); GlobalsBlog (2026)
Electrical wiring, cabling, conduit	20,000 – 80,000	[FACT] Pulse Energy (2026)
Electricity connection upgrade	50,000 – 2,00,000	[FACT] Pulse Energy (2026)
Civil works (trenching, mounting)	50,000 – 2,00,000	[FACT] Pulse Energy (2026)
Signage, bay marking, safety	10,000 – 30,000	[ESTIMATE]
Total per bay — low / mid / high	~1.7L / ~3.0L / ~6.0L	[ESTIMATE]

Table 5.1 — CAPEX per Charging Bay. Sources as cited. Equipment is only 30-50% of total investment; civil and grid works dominate (GlobalsBlog, 2026).

ANNUAL OPEX ITEM	COST (RS.)	STATUS
Electricity (Rs.6-15/kWh commercial tariff)	Variable per session	[FACT] Pulse Energy (2025)
Annual maintenance contract	15,000 – 50,000	[FACT] GlobalsBlog (2026)
Network / OCPP software (mandated for public stations)	24,000 – 1,20,000	[FACT] GlobalsBlog (2026)
Insurance	8,000 – 25,000	[FACT] GlobalsBlog (2026)
Fixed OpEx envelope per bay	~50,000 – 1,95,000	[ESTIMATE]

Table 5.2 — Operating Cost per Bay (excl. electricity). Sources as cited; the modelling in Sections 5.2-5.3 uses Rs.80,000 (reference) and Rs.70,000 (viable set) within this envelope.

5.2 Reference-Case Scenarios — Corrected

[ASSUMPTION] Reference inputs: 7.4 kW AC charger; consumer price Rs.12/kWh; 2-hour sessions delivering ~12 kWh; electricity Rs.9/kWh; host share 20% of gross; fixed OpEx Rs.80,000/bay/year. The table below is arithmetically exact.

METRIC	CONSERVATIVE	BASE	HIGH
Sessions / bay / day	3	6	10
kWh delivered / year	13,140	26,280	43,800
Gross revenue / year	Rs.1,57,680	Rs.3,15,360	Rs.5,25,600

METRIC	CONSERVATIVE	BASE	HIGH
Electricity @ Rs.9/kWh	Rs.1,18,260	Rs.2,36,520	Rs.3,94,200
Host share (20%)	Rs.31,536	Rs.63,072	Rs.1,05,120
Fixed OpEx / year	Rs.80,000	Rs.80,000	Rs.80,000
Net contribution / year	Rs.–72,116	Rs.–64,232	Rs.–53,720
Payback (Rs.3L bay)	Not achieved	Not achieved	Not achieved

Table 5.3 — Reference-Case Scenario Model (corrected arithmetic). Source: author's analysis from Pulse Energy (2025, 2026) and GlobalsBlog (2026) inputs. Illustrative scenarios, not forecasts.

The reference case fails structurally, not marginally. The operator retains $\text{Rs.}12 - \text{Rs.}9 - \text{Rs.}2.40 = \text{Rs.}0.60/\text{kWh}$, i.e. **Rs.7.20 per session**. Covering Rs.80,000 of fixed cost at Rs.7.20/session requires 11,111 sessions/year — **~30.4 sessions/bay/day**, roughly four times the physical maximum for 2-hour AC sessions. Higher utilisation barely helps: each added daily session contributes only ~Rs.2,628/year.

Reference-case verdict: not viable at any plausible utilisation.

Under Rs.12 / Rs.9 / 20% inputs, losses persist from 3 to 10 sessions/day and payback is never achieved. Any business plan using these inputs must be rejected or restructured — the required fix is not more volume but a wider per-kWh spread, derived next.

5.3 Sensitivity Analysis & the Viability Frontier

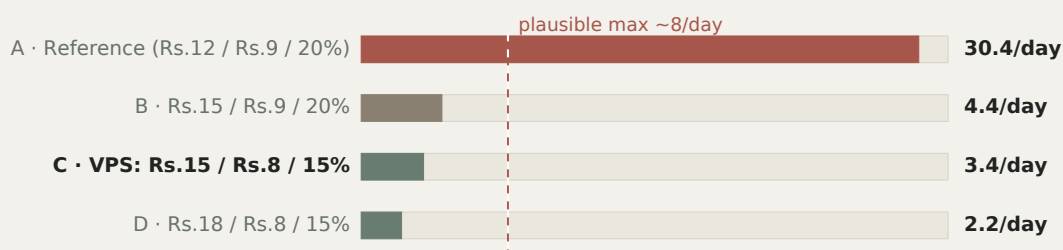
Because the reference case cannot break even, sensitivity is measured on the viable-parameter set (VPS) defined below — the envelope within which the model can work. Directional impacts on annual margin from the VPS base:

VARIABLE	CHANGE	IMPACT ON MARGIN
Sessions / day	6 → 4	Down ~Rs.42K
Sessions / day	6 → 8	Up ~Rs.42K
Electricity tariff	Rs.8 → Rs.11/kWh	Down ~Rs.79K — erases base margin
Consumer price	Rs.15 → Rs.13/kWh	Down ~Rs.45K
Host share	15% → 25%	Down ~Rs.39K
Charger CAPEX	Rs.3L → Rs.5L	Payback 5.5y → 9.1y
Charger CAPEX	Rs.3L → Rs.1.5L	Payback 5.5y → 2.7y
Charger uptime	90% → 75%	Down ~Rs.19K plus trust damage

Table 5.4 — Sensitivity of VPS Base-Case Margin. Source: author's analysis. Electricity tariff is the dominant risk: a Rs.3/kWh rise alone destroys viability.

The breakeven frontier below compares four price/tariff/share combinations. Only combinations holding breakeven below the physical ceiling (~8 sessions/day for AC destination bays) are investable:

Figure 5.2 — Breakeven Threshold (sessions/bay/day) Under Alternative Price / Tariff / Share Combinations



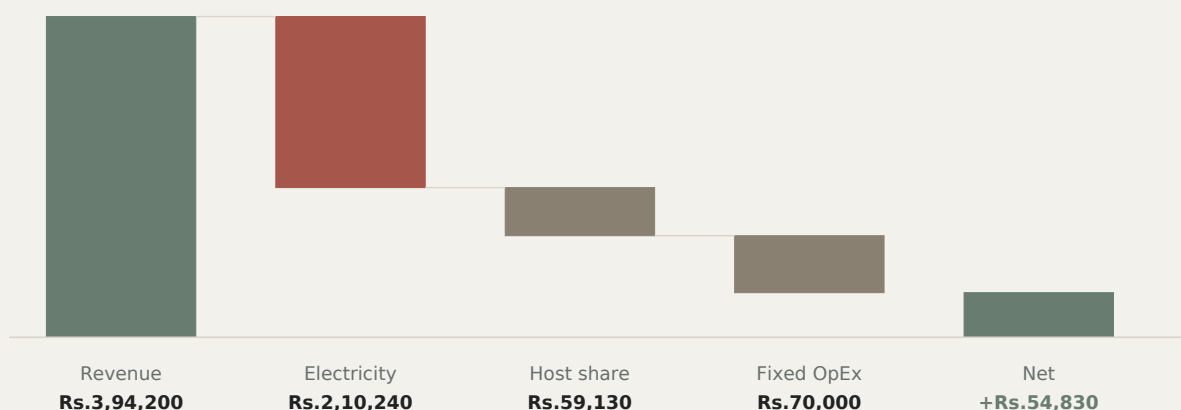
Source: author's analysis; fixed OpEx Rs.70,000/bay/year, 12 kWh/session. Combination A (reference) requires an impossible 30.4 sessions/day.

[ASSUMPTION] Viable-parameter set (VPS). Consumer price Rs.15/kWh (upper end of the observed Rs.8–15 AC range — a destination-convenience premium); electricity Rs.8/kWh (low-mid commercial tariff — a site-selection threshold); host share 15% of gross; fixed OpEx Rs.70,000/year (AMC Rs.25,000 + software Rs.30,000 + insurance Rs.15,000). Per-session margin: 12 kWh × Rs.4.75 = Rs.57.00; breakeven = 70,000 / (57 × 365) = **~3.4 sessions/bay/day**.

METRIC	CONSERVATIVE (3/DAY)	BASE (6/DAY)	HIGH (10/DAY)
kWh / year	13,140	26,280	43,800
Gross revenue / year	Rs.1,97,100	Rs.3,94,200	Rs.6,57,000
Electricity @ Rs.8/kWh	Rs.1,05,120	Rs.2,10,240	Rs.3,50,400
Host share (15%)	Rs.29,565	Rs.59,130	Rs.98,550
Fixed OpEx / year	Rs.70,000	Rs.70,000	Rs.70,000
Net contribution / year	Rs.–7,585	+Rs.54,830	+Rs.1,38,050
Payback (Rs.3L bay)	Not achieved	~5.5 years	~2.2 years

Table 5.5 — Viable-Parameter Scenario Model. Source: author's analysis. Do not treat the base case as a prediction; it is the threshold the pilot must clear.

Figure 5.1 — Annual Contribution Waterfall per Bay, Viable-Parameter Base Case (6 sessions/day)



Electricity remains the largest outflow (~53% of gross even in the VPS); the host share and fixed OpEx consume most of the remainder, leaving a thin but positive base margin. Source: author's model.

The VPS reframes the investment question precisely: the model is viable if and only if sites can be held inside the Rs.15/Rs.8/15% envelope while delivering ≥3.5 sessions/bay/day. Every element of

that sentence is testable — tariff by site survey, price by driver interviews, sessions by pilot.

5.4 Host Economics

The host weighs the revenue share against the bay's opportunity cost — a general parking space at a busy destination has real value:

HOST RECEIVES — MEASURABLE	HOST RECEIVES — UNPROVEN
— Revenue share: ~Rs.30K-Rs.99K/bay/year across VPS scenarios (Rs.59,130 base) [ESTIMATE] — Zero CAPEX and zero operational burden — Contracted income under hybrid base fee	— Incremental customer spend while charging [HYPOTHESIS] — requires POS-data experiment — EV-owner acquisition via green positioning [HYPOTHESIS] — untested in India

Critical assumption — host incentive.

At low utilisation the measurable share (~Rs.30K/year at 3 sessions/day) may not beat the parking opportunity cost at capacity-constrained sites. Durable host contracts therefore require either demonstrated volume or a hybrid base fee — both untested. Host economics, not driver demand alone, can veto the model.

The practical implication is contractual: pilot hosts should be offered the hybrid structure (base fee + reduced share) so the partnership survives the ramp-up period, with a contractual step-down to pure revenue share once the site clears 3.5 sessions/day for three consecutive months.

5.5 Three-Year Scenario Model (VPS)

SCENARIO MODEL — NOT A FORECAST. Directional illustration of network expansion under VPS base assumptions. Location counts, revenues and costs are modelling inputs dependent on unvalidated assumptions.

METRIC	YEAR 1	YEAR 2	YEAR 3
Active destination sites	5	15	35
Charging bays (avg 4/site)	20	60	140
Avg sessions/bay/day	3 (ramp-up)	5 (maturing)	6 (base)
Gross charging revenue	Rs.39L	Rs.1.97Cr	Rs.5.52Cr
Deployed CAPEX (cumulative)	Rs.60L	Rs.1.8Cr	Rs.4.2Cr
Annual contribution	Rs.-2L (est.)	+Rs.20L (est.)	+Rs.77L (est.)
Cumulative cash requirement	Rs.0.7Cr	Rs.2.2Cr	Rs.4.7Cr

Table 5.6 — Three-Year Directional Scenario (VPS). Source: author's model. L = lakh, Cr = crore. Cash requirement includes CAPEX, working capital and early losses; excludes corporate overhead, customer acquisition and platform development.

Key modelling observations. (1) Year 1 is mildly loss-making: ramp-up sessions sit below the 3.4/day breakeven. (2) Year-2 surplus depends on all 15 sites individually holding 5 sessions/day — portfolio averaging must not mask weak sites. (3) ~Rs.4.7 crore cumulative cash by Year 3 requires external capital and is sensitive to CAPEX overrun and tariff drift. (4) Excluded overheads (team, CAC, platform) would deepen early losses materially and must be modelled before fundraising.

CHAPTER 06

06 — Competitive & Strategic Analysis

6.1 Competitive Landscape

[FACT] India’s charging market is multi-layered: integrated utility CPOs, oil-marketing companies converting forecourts, private fast-charging specialists, OEM-affiliated networks, distributed peer-to-peer platforms and hardware enablers (Marqstats, 2026). Scale leaders by reported network size include Tata Power EZ Charge (5,500–8,000+ public points plus 86,000+ home chargers), Jio-bp pulse (6,000–7,000+ points), Statiq (7,000–8,000+), ChargeZone (2,200+ fast points across 58+ cities, highway-led) and Bolt.Earth (100,000+ distributed sockets across 1,900+ cities), with Ather Grid concentrated in two-wheelers (Electric Vehicle Insights, 2026; Autonexa, 2026; EVBlogs, 2025).

Two vectors threaten a destination entrant directly. First, large-balance-sheet players (Tata Power, Jio-bp) can outbid on host terms and lend brand credibility hosts trust. Second, asset-light partner models (Bolt.Earth, Statiq) already onboard hosts at scale through mechanics nearly identical to this study’s proposition — with existing app users and recognition. The window for independent site acquisition is finite and shrinking.

6.2 Competitive Comparison

OPERATOR	SCALE	FOCUS	HOST MODEL	RELEVANCE
Tata Power EZ Charge	Large	Highway + urban	Partner programmes	High
Jio-bp pulse	Large, growing	Fuel-adjacent, DC	Retail network	Medium
Statiq	Medium-Large	Urban, app-first	Active partner model	High
Bolt.Earth	Very large (P2P)	Residential, small commercial	Host model at scale	High
ChargeZone	Medium	Highway DC hubs	Limited destination	Medium
Ather Grid	Medium (2W)	Two-wheeler urban	2W host model	Low
Dest. entrant (this study)	Nascent	Destination-first AC	Core proposition	—

Table 6.1 — Competitive Landscape. Sources: Marqstats (2026), Electric Vehicle Insights (2026), Autonexa (2026), EVBlogs (2025), EVSelect (2026). Relevance ratings are qualitative judgements, not scored metrics.

The table’s uncomfortable message: no incumbent ignores destinations, and two (Statiq, Bolt.Earth) already execute host-partnership playbooks. Differentiation must therefore come from execution — site selection, uptime, host depth — not from the partnership concept itself, which is not proprietary.

6.3 Strategic Positioning

Defensible differentiation is possible but must be earned site by site:

ACHIEVABLE DIFFERENTIATION	REQUIRES EVIDENCE
— Underserved categories: mid-size offices, standalone hospitals, Tier-1 universities — Uptime leadership against the 18% national failure rate — Deep host integration: co-marketing, data sharing, branded experience — Fastest site onboarding via standardised install protocol	[HYPOTHESIS] Destination focus creates loyalty vs general-network loyalty [HYPOTHESIS] Enough underserved sites exist to reach sustainability before incumbents arrive [HYPOTHESIS] Host relationships resist switching to larger networks

The honest strategic summary: the entrant's edge is focus and speed against incumbents' scale and brand. That edge compounds only if early sites clear VPS thresholds quickly enough to fund the next wave before hosts sign with larger networks.

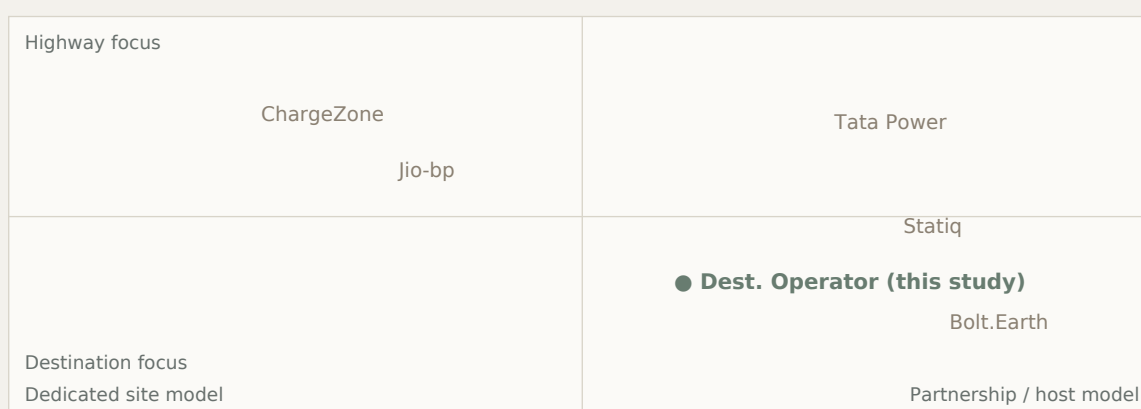
6.4 Location Strategy

[ASSUMPTION] Site selection is the operating decision that sets utilisation and hence economics. The criteria below structure evaluation; indicative thresholds must be calibrated with pilot data.

CRITERION	WHY IT MATTERS	INDICATIVE THRESHOLD
EV density in catchment	Thin density caps session volume	Top-20% registration density in city
Customer dwell time	Under ~45 min, kWh/session is uneconomic	>60 min average dwell
Parking headroom	Shortage creates EV-vs-general conflict	>15% inventory allocable to EV
Electricity infrastructure	Weak supply forces costly upgrades	3-phase supply; upgrade <Rs.1L/bay
Commercial tariff	Tariff above Rs.8 breaks the VPS	Effective tariff ≤ Rs.8/kWh
Competitor proximity	Adjacent chargers split sessions	No rival charger within ~500 m
Host commitment	Weak hosts churn at first friction	Minimum 12-month contract
Install accessibility	Complex layouts inflate civil works	Civil estimate <Rs.60K/bay

Table 6.2 — Site Selection Criteria. Source: author's analysis. Thresholds are indicative assumptions pending pilot calibration.

Figure 6.1 — Destination-Charging Operator Positioning Map (indicative, qualitative)



Source: author's qualitative positioning — illustrative, not metric-scored. The entrant occupies the partnership-led, destination-focused quadrant contested most directly by Statiq and Bolt.Earth.

CHAPTER 07

07 — Validation & Risk Analysis

7.1 Key Hypotheses Requiring Validation

Six hypotheses carry the model. None has been tested. Each maps to evidence, method and the consequence of failure:

ID	HYPOTHESIS	EVIDENCE REQUIRED	METHOD	IF FALSE
H1	Drivers regularly use destination charging on routine visits	Stated preference + session frequency	Driver interviews + pilot logs	Utilisation below breakeven
H2	Hosts allocate bays on 12-month terms	Surveys + signed agreements	Host interviews + negotiation	Network cannot form
H3	Charging lifts host customer spend measurably	EV vs non-EV transaction data	POS-data pilot experiment	Share alone must justify hosts
H4	Sites reach ≥ 3.5 sessions/bay/day in 6–12 months	Session logs per site	Pilot, ≥ 3 months data	Model stays loss-making
H5	15% host share secures and retains hosts	Outcomes across 5+ categories	Commercial negotiation data	Churn (too low) or margin loss (too high)
H6	$\geq 90\%$ uptime is maintainable at destination sites	Uptime logs per site	Pilot monitoring, monthly	Distrust, session loss, host damage

Table 7.1 — Hypotheses and Validation Requirements. Source: author’s analysis. H4’s threshold follows the VPS breakeven (3.4/day); the reference case has no achievable threshold.

7.2 Validation Plan

PHASE 1 — DISCOVERY (WEEKS 1-8)

Driver interviews (n=20-30) across 2 cities: charging behaviour, home access, visit frequency, price response. Tests H1, informs H4.

Host interviews (n=15-20) with mall, hotel, hospital, office managers: parking pressure, bay willingness, share expectations. Tests H2, H5.

PHASE 2 — COMMERCIAL (WEEKS 8-16)

Host intent signal: operator landing page measuring enquiry-to-meeting conversion. Early H2 evidence without deployment.

Site economics: per-site models for 8-10 prospects using local tariffs, registration density and 2-3 installation quotes each.

PHASE 3 — PILOT (MONTHS 4-10)

3-5 heterogeneous pilots (mall, hotel, hospital/university, office), 3-4 bays each: sessions, kWh, demographics, uptime, host satisfaction. Tests H1, H4, H6.

H3 experiment (optional): POS-data comparison of EV vs non-EV spend with host consent; privacy-reviewed.

PHASE 4 — DECISION (MONTHS 10-12)

Go / No-Go / Pivot against pre-registered thresholds: ≥ 3 of 5 sites at ≥ 3.5 sessions/day by month 6, zero host churn, $\geq 90\%$ uptime.

Locked metrics: sessions/bay/day, uptime, renewal rate, CAC, revenue/site, pilot NPV — thresholds fixed before the pilot, never adjusted ex-post.

Figure 7.1 — Four-Phase Validation Plan. Source: author's framework. Total direct pilot cost is bounded by 12-20 bays of CAPEX plus operating losses — the cheapest reliable answer to the viability question.

7.3 Risk Matrix

Risks are rated qualitatively on probability $\times$ impact from industry evidence and the Chapter-5 model — not actuarially:

RISK	PROB.	IMPACT	RATING	MITIGATION
Persistent low utilisation (<3/day at maturity)	Med-High	Critical	HIGH	Strict site criteria; exit clauses for weak sites
Incumbent competition for the same hosts	High	High	HIGH	First-mover contracts; exclusivity; faster onboarding
Host churn at renewal	Medium	High	MED-HIGH	Long terms; hybrid fees; engagement programme
Charger downtime (<85% uptime)	Medium	High	MED-HIGH	Preventive maintenance; certified hardware; NOC monitoring
Tariff volatility (>20% rise)	Medium	High	MED-HIGH	Tariff-indexed consumer pricing; site tariff caps
ICE-blocking of EV bays	High	Medium	MEDIUM	Bollards; enforcement protocol; bay monitoring
CAPEX overrun	Medium	Medium	MEDIUM	Standard installs; pre-surveyed capacity; contingency
Grid / regulatory delay	Medium	Low-Med	LOW-MED	Prioritise adequate-capacity sites; buffer lead times
Customer acquisition cost	Medium	Low-Med	LOW-MED	App interoperability (OCPI); host co-marketing

Table 7.2 — Risk Register. Source: author's analysis. Tariff risk is rated up versus conventional assessments because Table 5.4 shows a Rs.3/kWh rise erases the entire VPS margin.

	HIGH PROB.	MEDIUM PROB.	LOW PROB.
HIGH IMPACT	Low utilisation Incumbent competition	Host churn Downtime Tariff volatility	
MEDIUM IMPACT	ICE-blocking	CAPEX overrun	CAC

LOW IMPACT		Grid / regulatory	
------------	--	-------------------	--

Figure 7.2 — Risk Probability × Impact Matrix. Source: author’s analysis. The red zone — utilisation, competition, hosts, downtime, tariffs — is the pilot’s measurement agenda.

CHAPTER 08

08 — Findings & Conclusion

What Is Supported by Evidence

EV MARKET GROWTH

Passenger EVs +77% in CY2025; 2.27M total EVs. Demand for charging will rise. **[FACT]**

INFRASTRUCTURE GAP

1:235 charger-to-EV ratio; 1.32M stations needed by 2030. The gap is structural. **[FACT]**

LAND-COST PROBLEM

Urban leases absorb up to 40% of dedicated-project budgets. Host access avoids this. **[FACT]**

POLICY ALIGNMENT

De-licensed activity; Rs.2,000Cr PM E-DRIVE; designated host categories. **[FACT]**

What Is Directionally Supported

Under viable parameters (Rs.15 / Rs.8 / 15%), base-case unit economics are positive: +Rs.54,830/bay/year at 6 sessions/day, ~5.5-year payback.

Directionally viable — but conditional on holding every VPS input simultaneously. A single-parameter slip (tariff Rs.8→Rs.11) erases the margin. The pilot must prove the envelope, not just the volume.

The destination model structurally reduces site cost versus dedicated stations.

Avoided land acquisition and shared electrical infrastructure lower CAPEX per bay — though the magnitude depends on site-specific upgrade needs, which vary 4× in the data.

What Remains Unproven

UNPROVEN ELEMENT	WHY	IF WRONG
Driver demand for destination charging	No interviews or session data	Sessions below breakeven; model fails
Host willingness to commit parking	No negotiations conducted	Network cannot form
Incremental host revenue from EV users	No transaction data	Share alone must justify hosts
Sustainable competitive position	Incumbents expanding into hosts	First-mover window closes early
VPS tariff/price envelope	Rs.15/Rs.8/15% jointly untested	Reverts to reference-case losses

Table 8.1 — Unproven Elements. Source: author's analysis. Any single row failing is sufficient to reject or pivot the model.

Conditions for Viability

01 — UTILISATION

≥3.5 sessions/bay/day within 6–12 months. Below this the VPS base does not contribute.

02 — PRICE / TARIFF SPREAD

Consumer price ≥ Rs.15 with commercial tariff ≤ Rs.8. The spread is the business.

03 — HOST SHARE ≤ 15-20%

Higher shares transfer the margin to the host. Hybrid fees only during ramp-up.

04 — UPTIME ≥ 90%

The 18% national failure rate must not be replicated; trust is the product.

05 — COMPETITIVE ACCESS

Secure hosts before larger networks do. The window is finite and shrinking.

06 — SITE DISCIPLINE

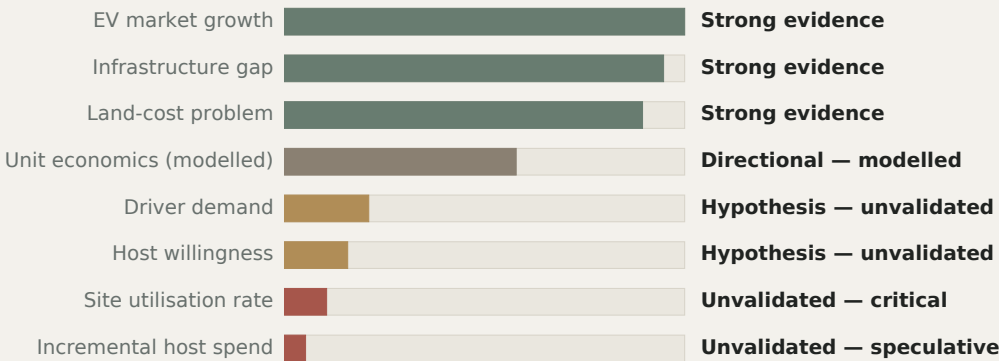
Reject sub-threshold sites; exit weak ones fast. Averages must not hide failures.

Recommendation

The destination-charging model is structurally coherent and addresses a documented infrastructure gap. Its economics are viable **only inside a narrow, testable parameter envelope** — and every parameter in that envelope is currently unvalidated. **No scaling decision should be made on the basis of this analysis alone.**

The recommended next step is the structured, time-bounded validation programme in Chapter 7 (Phases 1-3), designed to force a Go / No-Go / Pivot decision within 10-12 months against pre-registered thresholds that are never adjusted after data collection.

Figure 8.1 — Evidence Maturity by Business Model Element



Source: author’s assessment. Bar length reflects qualitative confidence, not a statistical measure. The drop from market evidence to site-level validation is the report’s central message.

Research limitations. This study relies entirely on secondary data, industry reports and analytical modelling. No primary interviews, negotiations or pilot data were collected. All models contain unvalidated assumptions. Use this report as the foundation for a primary-research programme — not as a basis for capital allocation.

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APPENDIX

Appendix

A — Customer Interview Questions

1. Where do you currently charge your EV most often, and why that location?
2. Do you have access to reliable home or workplace charging? If not, what blocks it?
3. Which destinations (mall, office, hotel, hospital, other) do you visit weekly, and how long do you typically stay?
4. Have you ever wanted to charge at a destination but could not? What happened?
5. What would you pay per kWh for charging during shopping or dining, versus your home tariff?
6. How do you find chargers today, and what frustrates you most about the experience?
7. Have you encountered non-working public chargers? How did it affect your trust?
8. Would guaranteed bay availability (enforced EV-only bays) change your charging habits?
9. What payment method would you prefer — existing UPI apps, RFID, or a dedicated app?
10. What would stop you from using destination charging even if it were available nearby?

B — Host Interview Questions

1. How many parking bays do you operate, and what is typical occupancy by day and hour?
2. Have EV drivers ever asked for charging at your location? How often?
3. Would you allocate 2-4 bays to EV charging under a 12-month agreement? What concerns you?
4. What revenue share or monthly fee would make allocated bays worthwhile versus general parking?
5. Would a guaranteed base fee change your willingness? At what level?
6. Is 3-phase supply available near the parking area? Has load capacity ever constrained you?
7. Who would enforce EV-only bay usage, and what enforcement tools exist today?
8. Would you share anonymised footfall or transaction data for a charging-impact study?
9. What would make you terminate a charging partnership at renewal?
10. Which operator attributes matter most — brand, revenue, reliability, or marketing support?

C — Unit-Economics Assumptions Log

Every modelling input in Chapter 5, with source and validation route:

#	INPUT	VALUE USED	SOURCE / STATUS	VALIDATED BY
C1	Charger capacity	7.4 kW AC	[ASSUMPTION]	Procurement spec
C2	Energy per session	12 kWh (2 hrs)	[ASSUMPTION]	Pilot meter data
C3	Consumer price (ref / VPS)	Rs.12 / Rs.15 per kWh	Range [FACT] ; VPS [ASSUMPTION]	Driver interviews
C4	Electricity tariff (ref / VPS)	Rs.9 / Rs.8 per kWh	Range [FACT] ; VPS [ASSUMPTION]	Site survey
C5	Host share (ref / VPS)	20% / 15% of gross	[ASSUMPTION]	Host negotiation
C6	Fixed OpEx (ref / VPS)	Rs.80,000 / Rs.70,000 per year	Envelope [FACT]	Vendor quotes
C7	CAPEX per bay (mid)	Rs.3,00,000	[ESTIMATE]	Installation quotes

#	INPUT	VALUE USED	SOURCE / STATUS	VALIDATED BY
C8	Home-charging gap	40–55% of e-PV owners	[ASSUMPTION]	Driver interviews
C9	Dwell times by category	Table 4.1 values	[ESTIMATE]	Host footfall data
C10	Uptime target	≥90%	[HYPOTHESIS]	Pilot logs

D — Pilot Metrics & Decision Thresholds

Measured monthly per site; Go / No-Go evaluated at month 6 against thresholds registered before launch:

METRIC	DEFINITION	GO THRESHOLD	NO-GO
Sessions / bay / day	Completed sessions ÷ bays ÷ days	≥3.5 on ≥3 of 5 sites	<2.5 on majority
kWh per session	Metered energy ÷ sessions	≥10 kWh mean	<7 kWh mean
Charger uptime	Available hours ÷ total hours	≥90%	<80%
Host renewal intent	Hosts renewing at month 6	100% renew	Any churn
Customer acquisition cost	Marketing ÷ active users	<Rs.500	>Rs.1,500
Gross revenue per site	Monthly kWh × realised price	Covers site opex + share	<70% of opex
Net promoter (drivers)	Recommend-to-other-drivers score	≥ +30	Negative
ICE-blocking rate	Blocked bay-hours ÷ total	<5%	>15%

Pivot triggers: if utilisation clears thresholds only at offices/hotels, narrow the model to long-dwell categories; if tariff discipline fails, test fixed-fee host contracts; if price resistance appears at Rs.15, re-derive the frontier before continuing. Any threshold change after data collection invalidates the decision.